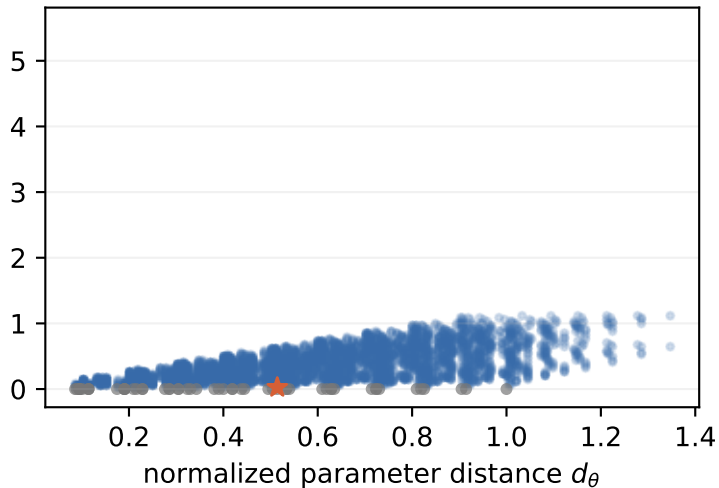
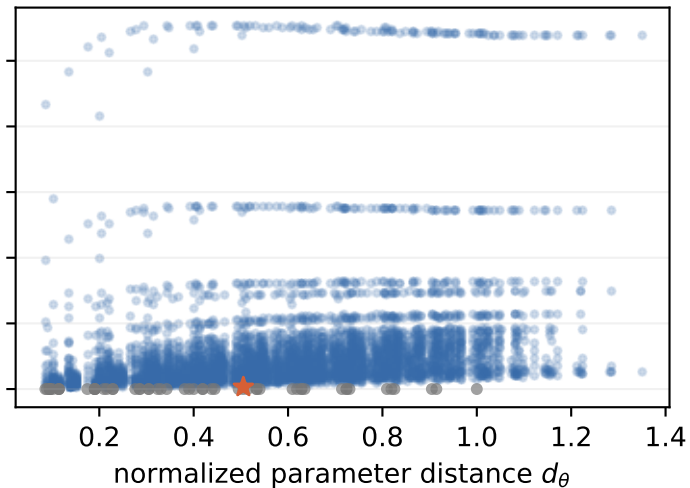


standardized RMS observable distance

Ringdown



Ringdown + photon geometry



• finite-k pairs • k=0 null control ★ closest pair, $d_\theta \geq 0.5$